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Unit – 1 : Session – 8 : SLO – 2 | Binary arithmetic and BCD operations

■ I. Match the Following

1. Sign bit in binary number — b) 1 for negative and 0 for positive
2. 1's complement of 0101 — a) 1010
3. 2's complement of 1001 — e) 0111 (0110 + 1)
4. Binary division rule — c) Shift and subtract process
5. BCD for decimal 42 — d) 0100 0010

■ II. Fill in the Blanks

1. 0
2. 1101
3. 1
4. decimal long
5. 4

■ III. Activity / Describe and Perform

1. Sign Magnitude — $(+4)+(-2)$: different signs $\rightarrow 100-010 = 010$, sign of larger (+) $\rightarrow 0010 = +2$. $(-4)-(+2) = -(4+2) = -6 = 1110$.
2. 1's complement subtraction, $11-5$: $1011 + 1$'s of 0101 (1010) = $1\ 0101 \rightarrow$ end-around carry $\rightarrow 0101+1 = 0110 = +6$.
3. 2's complement subtraction, $6-3$: $0110 + 2$'s of 0011 (1101) = $1\ 0011 \rightarrow$ discard carry $\rightarrow 0011 = +3$.
4. Multiplication 110×11 (6×3): $110 + 1100$ (shifted) = $10010 = 18$. Division $1100 \div 11$ ($12 \div 3$): long division $\rightarrow$ quotient $0100 = 4$, remainder 0.
5. BCD: $42 = 0100\ 0010$. BCD addition $27+35$: $0010\ 0111 + 0011\ 0101 = 0101\ 1100$; units 1100 invalid $\rightarrow +0110 \rightarrow 0110\ 0010$ with carry $\rightarrow = 62$ ($0110\ 0010$).